Chengshuai Zhao

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Education

Arizona State University — Jan 2024 – May 2028 (expected)
Ph.D. in Computer Science, Advisor: Huan Liu — *Tempe, AZ, USA*

University of California, Irvine — Sep 2022 – Jun 2023
B.S. in Data Science — *Irvine, CA, USA*

Huazhong Agricultural University — Sep 2018 – Jun 2023
B.S. in Data Science · B.S. in Life Science — *Wuhan, China*

Research Experience

Data Mining and Machine Learning Lab (DMML), Arizona State University — Jan 2024 – Present
Graduate Research Assistant, advised by Prof. Huan Liu — *Tempe, AZ, USA*

- **LLM post-training & reasoning:** reasoning distillation via explanatory data synthesis and RL with structured rewards [4]; an RL-optimized knowledge graph retriever for reasoning-augmented generation [12]; a controlled and synthetic environment for pre-training LLMs from scratch and probing reasoning [7].
- **LLM/MLLM system security, privacy & safety:** bilevel-optimized knowledge-poisoning attacks on RAG systems [19] and covert malicious-agent injection into MLLM societies [18]; adversarial data synthesis that protects data from unauthorized fine-tuning of LLMs [30] and VLMs [31]; RL-driven defenses that shield proprietary LLMs from unauthorized API-based knowledge distillation [32].
- **RAG & LLM agent applications:** knowledge-graph-grounded response verification in RAG for domain-specific question answering [13], [14]; a multi-agent RL framework in Graph-RAG for multi-hop reasoning [12]; LLM multi-agent frameworks with human-in-the-loop collaboration for large-scale content analysis [15], [16].
- **LLM evaluation:** LLM-as-a-judge [10] and the detectability of LLM-generated judgments [11]; detection and evaluation of LLM-generated content [17], [33]; evaluation of CoT reasoning under distribution shift [7].
- **Causality, interpretability, efficiency & recommendation:** causality-guided representation learning for robust generalization across domains, styles, and modalities [3], [8], [28]; concept bottleneck models for interpretable and trustworthy open-ended grading [9], [1]; low-rank node-level graph diffusion for few-shot node classification [6]; diffusion language models for diverse and efficient bundle recommendation [27].

Center for AI Research and Innovation (CAIRI), Westlake University — Jul 2022 – Sep 2022
Research Intern, advised by Prof. Stan Z. Li — *Hangzhou, China*

- **Molecular foundation models:** self-supervised pre-training of GNNs for molecules with a context-aware atom-level tokenizer and masked atom modeling with contrastive learning [20].

BioMedical Big Data Mining Lab (BBDM), Huazhong Agricultural University — Oct 2019 – Sep 2022
Undergraduate Researcher, advised by Prof. Wen Zhang — *Wuhan, China*

- **Graph representation learning & link prediction:** contrastive self-supervised GNNs, equivariant positional-encoding GNNs, and graph-embedding ensembles for predicting molecular interactions [21], [23], [24].

Publications

Google Scholar | Citations: **1,702** | h-index: **12** | i10-index: **14** * Equal contribution

Conference, Journal & Workshop Papers

[1] K. S. Chaudhary, **C. Zhao**, F. Zhang, G. Agrawal, Y. Deng, and H. Liu, "EssayCBM: Rubric-Aligned Concept Bottleneck Models for Transparent Essay Grading," **ECML-PKDD 2026**.

[2] Y. Li, **C. Zhao**, G. Agrawal, Y. Deng, J. Park, Y.-C. Chen, and H. Liu, "Epistemic Cognition and Uncertainty Navigation with a Domain-Specific AI Chatbot in STEM Education," **LERN 2026**.

[3] P. Ma, **C. Zhao**, B. Jiang, S. Vishnubhatla, U. Jeong, A. Beigi, A. Raglin, and H. Liu, "CAMO: Causality-Guided Adversarial Multimodal DOmain Generalization for Crisis Classification," **PAKDD 2026**. ★ **Oral**.

[4] Z. Tan, **C. Zhao**, S. Wang, J. Li, T. Chen, and H. Liu, "Probing to Refine: Reinforcement Distillation of LLM Reasoners via Explanatory Inversion," **ICLR 2026**.

[5] S. Wang, Z. Tan, Z. Chen, D. Li, Y. Zhu, B. Jiang, Y. He, **C. Zhao**, Z. Lei, P. Sheth, L. Li, L. P.-Y. Ting, J. Li, and H. Liu, "Large Language Models for Data Science: A Survey," **SIGKDD Explorations, 2026**.

[6] Y. Wang, **C. Zhao**, D. Sun, H. Liu, and Y. Yang, "Low-Rank Few-Shot Node Classification by Node-Level Graph Diffusion," **ICLR 2026**.

[7] **C. Zhao**, Z. Tan, P. Ma, D. Li, B. Jiang, Y. Wang, Y. Yang, and H. Liu, "Is Chain-of-Thought Reasoning of LLMs a Mirage? A Data Distribution Lens," **ACL 2026**.

[8] **C. Zhao**, S. Wan, P. Sheth, K. Patwa, K. S. Candan, and H. Liu, "Causality Guided Representation Learning for Cross-Style Hate Speech Detection," **WWW 2026**. ★ **Oral**.

[9] **C. Zhao**, F. Zhang, K. S. Chaudhary, Y. Li, L. P.-Y. Ting, Y.-C. Chen, and H. Liu, "REC-CBM: Rubric-Aware Error-Correction Concept Bottleneck Models for Trustworthy Open-Ended Grading," **SDM 2026**. 🏆 **Best Applied Data Science Paper Award**.

[10] D. Li, B. Jiang, L. Huang, A. Beigi, **C. Zhao**, Z. Tan, A. Bhattacharjee, Y. Jiang, C. Chen, T. Wu, K. Shu, L. Cheng, and H. Liu, "From Generation to Judgment: Opportunities and Challenges of LLM-as-a-judge," **EMNLP 2025**. ★ **Oral**.

[11] D. Li, Z. Tan, **C. Zhao**, B. Jiang, B. Huang, P. Ma, A. Alnaibari, K. Shu, and H. Liu, "Who's Your Judge? On the Detectability of LLM-Generated Judgments," **Lock-LLM @ NeurIPS 2025**.

[12] L. P.-Y. Ting*, **C. Zhao***, Y.-H. Zeng, Y. J. Lim, K.-T. Chuang, and H. Liu, "Leaps Beyond the Seen: Reinforced Reasoning Augmented Generation for Clinical Notes," **GenAI4Health @ NeurIPS 2025**.

[13] **C. Zhao**, G. Agrawal, F. Zhang, T. Kumarage, Z. Tan, Y. Deng, Y. Chen, and H. Liu, "Ontology-Aware RAG for Improved Question-Answering in Cybersecurity Education," **IEEE BigData 2025**.

[14] **C. Zhao**, R. D. Maria, T. Kumarage, K. S. Chaudhary, G. Agrawal, Y. Li, J. Park, Y. Chen, Y. Deng, and H. Liu, "CyberBOT: Ontology-Grounded Retrieval Augmented Generation for Reliable Cybersecurity Education," **CIKM 2025**. 🏆 **Best Demo Paper Award**.

[15] **C. Zhao**, Z. Tan, C. Wong, X. Zhao, T. Chen, and H. Liu, "SCALE: Towards Collaborative Content Analysis in Social Science with Large Language Model Agents and Human Intervention," **ACL 2025**. ★ **Oral**.

[16] X. Zhao, **C. Zhao**, M. Mengesteab, C.-W. Wong, M. Zhan, Z. Tan, and T. Chen, "Automating Content Analysis With Multiple LLM Agents: Impacts of Agent Attributes and Human–AI Collaboration," **Social Science Computer Review, 2025**.

[17] B. Jiang*, **C. Zhao***, Z. Tan, and H. Liu, "Catching Chameleons: Detecting Evolving Disinformation Generated using Large Language Models," **IEEE CogMI 2024**. 🏆 **Best Student Paper Award**.

[18] Z. Tan*, **C. Zhao***, R. Moraffah, Y. Li, Y. Kong, T. Chen, and H. Liu, "The Wolf Within: Covert Injection of Malice into MLLM Societies via an MLLM Operative," **ReGenAI @ CVPR 2024**.

[19] Z. Tan*, **C. Zhao***, R. Moraffah, Y. Li, S. Wang, J. Li, T. Chen, and H. Liu, "Glue pizza and eat rocks - Exploiting Vulnerabilities in Retrieval-Augmented Generative Models," **EMNLP 2024**.

[20] J. Xia*, **C. Zhao***, B. Hu, Z. Gao, C. Tan, Y. Liu, S. Li, and S. Z. Li, "Mole-BERT: Rethinking Pre-training Graph Neural Networks for Molecules," **ICLR 2023**.

[21] **C. Zhao**, H. Wang, W. Qi, and S. Liu, "Toward drug-miRNA resistance association prediction by positional encoding graph neural network and multi-channel neural network," **Methods, 2022**.

[22] X. Xu, S. Liu, Z. Yang, X. Zhao, Y. Deng, G. Zhang, J. Pang, **C. Zhao**, and W. Zhang, "A systematic review of computational methods for predicting long noncoding RNAs," **Briefings in Functional Genomics, 2021**.

[23] **C. Zhao**, S. Liu, F. Huang, S. Liu, and W. Zhang, "CSGNN: Contrastive Self-Supervised Graph Neural Network for Molecular Interaction Prediction," **IJCAI 2021**.

[24] **C. Zhao**, Y. Qiu, S. Zhou, S. Liu, W. Zhang, and Y. Niu, "Graph embedding ensemble methods based on the heterogeneous network for lncRNA-miRNA interaction prediction," **BMC Genomics, 2020**.

Preprints & Papers Under Review

[25] C.-H. Kuo*, **C. Zhao***, H. Liu, and L. P.-Y. Ting, "Self-Supervised Latent Action-Conditioned World Modeling for Time Series," *Under review at WWW 2027*.

[26] Z. Liu, **C. Zhao**, and H. Liu, "MUGEN: Generating Unlearnable Graph Examples for Multiple Learning Tasks," *Under review at WSDM 2027*.

[27] P. Ma, B. Jiang, **C. Zhao**, D. Li, Z. Tan, and H. Liu, "Diversity-Preserving and Efficient Diffusion Language Model Recommendation for Bundle Construction," *Under review at KDD 2027*.

[28] P. Ma, **C. Zhao**, U. Byahut, B. Jiang, S. Vishnubhatla, U. Jeong, A. Raglin, A. Harrison, and H. Liu, "CaDiF: Few-Shot Multimodal Learning via Causality-Based Disentanglement and Cross-Modal Flattening," *Under review at WSDM 2027*.

[29] K. Wei, X. Shen, **C. Zhao**, T. Chen, S. Wang, and Z. Tan, "WISE: World-Model-Guided Interaction and Selective Knowledge Evolution for Self-Improving Agents," *Under review at NAACL 2027*.

[30] **C. Zhao**, P. Ma, D. Li, B. Jiang, Z. Yu, Z. Tan, and H. Liu, "TextCloak: Thwarting Unauthorized LLM Exploitation via RL-Driven Unlearnable Text," *Under review at AAAI 2027*.

[31] **C. Zhao**, Z. Tan, D. Li, Z. Yu, and H. Liu, "To See is Not to Learn: Protecting Multimodal Data from Unauthorized Fine-Tuning of Large Vision-Language Model," *Under review at NDSS 2027*.

[32] **C. Zhao**, F. Zhang, D. Li, B. Jiang, P. Ma, Z. Yu, Z. Tan, and H. Liu, "Rethinking Anti-distillation of Large Language Models: An Unlearnable Examples Perspective," *Under review at ICLR 2027*.

[33] B. Jiang, D. Li, Z. Tan, **C. Zhao**, and H. Liu, "Are Today's LLMs Ready to Explain Well-Being Concepts?" *Under review at AAAI 2027*.

Honors & Awards

Awards

- **Outstanding Research Award** — Nov 2026
  Arizona State University — *Tempe, AZ, USA*
- **Outstanding Mentor Award** — Nov 2026
  Arizona State University — *Tempe, AZ, USA*
- **Best Applied Data Science Paper Award** — Oct 2026
  SIAM International Conference on Data Mining (SDM 2026) — *Salt Lake City, UT, USA*
- **Gold Reviewer Award** — May 2026
  The 43rd International Conference on Machine Learning (ICML 2026) — *Seoul, Korea*

- **Best Demo Paper Award** Nov 2025
  The 34th ACM International Conference on Information and Knowledge Management (CIKM 2025) *Seoul, Korea*
- **Best Student Paper Award** Oct 2024
  The 6th IEEE International Conference on Cognitive Machine Intelligence (CogMI 2024) *Washington, DC, USA*

Research Grants & Travel Awards

- **JumpStart Research Grant** Oct 2026
  Arizona State University *Tempe, AZ, USA*
- **Student Travel Award** Sep 2026
  SIAM International Conference on Data Mining (SDM 2026) *Salt Lake City, UT, USA*
- **SCAI Conference Travel Award** May 2026
  School of Computing and Augmented Intelligence, Arizona State University *Tempe, AZ, USA*

Invited Talks

- *Invited Talk*, NICE AI Talk, Virtual Dec 2025
- *Invited Talk*, CCSB Biweekly Seminar Series, Prairie View A&M University, Virtual Sep 2025
- *Guest Lecture*, CSE 472: Social Media Mining, Arizona State University, Tempe, AZ, USA Aug 2025
- *Invited Talk*, Yochan Lab, Arizona State University, Tempe, AZ, USA Aug 2025

Media Coverage

- *The New Yorker*: "What if A.I. Doesn't Get Much Better Than This?" Aug 2025
- *Ars Technica*: "LLMs' 'Simulated Reasoning' Abilities Are a 'Brittle Mirage,' Researchers Find" Aug 2025
- *VentureBeat*: "LLMs Generate 'Fluent Nonsense' When Reasoning Outside Their Training Zone" Aug 2025
- *SF Examiner*: "New Study Calls AI Reasoning a 'Brittle Mirage'" Aug 2025
- *The Decoder*: "Yet Another Study Doubts That LLM Reasoning Shows True Logic over Pattern Imitation" Aug 2025

Professional Service

Memberships

- IEEE · ACM · AAAI · SIAM · AAAS

Program Committee Member / Reviewer

- **Data Mining & Information Retrieval**: SIGKDD, WWW, SIGIR, WSDM, CIKM, ICDM 2024 – Present
- **Machine Learning & AI**: ICML, NeurIPS, ICLR, AAAI, IJCAI 2025 – Present
- **Natural Language Processing**: ACL, EMNLP, NAACL, EACL 2024 – Present

Workshop Organization

- *Organizer & Speaker*, Workshop on Cybersecurity, AI, and Education, 2026 SFS PI Meeting and Symposium, Arlington, VA, USA Feb 2026

Technical Skills

- **Research Skills**: LLM post-training (supervised fine-tuning, reinforcement learning, knowledge distillation), LLM reasoning, security, privacy & safety of LLM and multimodal LLM systems, LLM agents, retrieval-augmented generation (RAG), LLM evaluation, explainable AI & model interpretability, world models, diffusion models,

graph mining (link prediction, node classification, graph generation), causal representation learning, time-series analysis, recommender systems, AI for science (biomedical, healthcare, social science, education)

- **Programming Languages**: C, C++, Java, Python, R, Go, Scala, SQL, Verilog
- **Frameworks**: PyTorch, Hugging Face Transformers, DeepSpeed, vLLM, TensorFlow, scikit-learn, Hadoop, Spark
- **Databases**: Oracle, MySQL, SQLite, MongoDB
- **Tools**: Git, Docker, Conda, Overleaf, Claude Code
- **Operating Systems**: Ubuntu, CentOS, macOS, Windows